```
/* Distance Vector Routing in this program is implemented using Bellman Ford Algorithm:-
*/
#include<stdio.h>
struct node {
    unsigned dist[20];
    unsigned from[20];
} rt[10];
int main() {
    int costmat[20][20];
    int nodes,i,j,k,count=0;
    printf("Enter the number of nodes : ");
    scanf("%d",&nodes);//Enter the nodes
    printf("Enter the cost matrix :\n");
    for(i=0;i<nodes;i++) {
        for(j=0;j<nodes;j++) {
            scanf("%d",&costmat[i][j]);
            costmat[i][i]=0;
            rt[i].dist[j]=costmat[i][j];//initialise the distance equal to cost matrix
            rt[i].from[j]=j;
        }
    }
    do {
        count=0;
        for(i=0;i<nodes;i++)//We choose arbitary vertex k and we calculate the direct
        distance from the node i to k using the cost matrix
            //and add the distance from k to node j
            for(j=0;j<nodes;j++)
                for(k=0;k<nodes;k++)
                    if(rt[i].dist[j]>costmat[i][k]+rt[k].dist[j]) {
                        //We calculate the minimum distance
                        rt[i].dist[j]=rt[i].dist[k]+rt[k].dist[j];
                        rt[i].from[j]=k;
                        count++;
                    }
    } while(count!=0);
    for(i=0;i<nodes;i++) {
        printf("For router %d\n",i+1);
        for(j=0;j<nodes;j++) {
            printf("\tnode %d via %d Distance %d \n",j+1,rt[i].from[j]+1,rt[i].dist[j]);
        }
    }
    printf("\n");
    getch();
}

/*
A sample run of the program works as:-
Enter the number of nodes :
3
Enter the cost matrix :
0 2 7
2 0 1
7 1 0
For router 1
node 1 via 1 Distance 0
node 2 via 2 Distance 2
node 3 via 3 Distance 3
For router 2
node 1 via 1 Distance 2
node 2 via 2 Distance 0
node 3 via 3 Distance 1
For router 3
node 1 via 1 Distance 3
node 2 via 2 Distance 1
node 3 via 3 Distance 0
*/

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// Exp 11 : Write a program to implement Link state routing
```

```c
#include <stdio.h>
int main() {
    int count,src_router,i,j,k,w,v,min;
    int cost_matrix[100][100],dist[100],last[100];
    int flag[100];
    printf("Enter the no of routers");
    scanf("%d",&count);
    printf("Enter the cost matrix values:\n");
    for(i=0;i<count;i++) {
        for(j=0;j<count;j++) {
            printf("%d->%d:",i,j);
            scanf("%d",&cost_matrix[i][j]);
            if(cost_matrix[i][j]<0)cost_matrix[i][j]=1000;
        }
    }
    printf("Enter the source router:");
    scanf("%d",&src_router);
    for(v=0;v<count;v++) {
        flag[v]=0;
        last[v]=src_router;
        dist[v]=cost_matrix[src_router][v];
    }
    flag[src_router]=1;
    for(i=0;i<count;i++) {
        min=1000;
        for(w=0;w<count;w++) {
            if(!flag[w])
                if(dist[w]<min) {
                    v=w;
                    min=dist[w];
                }
        }
        flag[v]=1;
        for(w=0;w<count;w++) {
            if(!flag[w])
                if(min+cost_matrix[v][w]<dist[w]) {
                    dist[w]=min+cost_matrix[v][w];
                    last[w]=v;
                }
        }
    }
    for(i=0;i<count;i++) {
        printf("\n%d==>%d:Path taken:%d",src_router,i,i);
        w=i;
        while(w!=src_router) {
            printf("<--%d",last[w]);
            w=last[w];
        }
        printf("\tShortest path cost:%d",dist[i]);
    }
    return 0;
}
```

```c
// Exp 12 :: Write a program for congestion control using Leaky bucket algorithm
#include <stdio.h>
#include <stdbool.h>
void leakyBucket(int inputRate, int bucketSize, int packetSize, int totalTime) {
    int bucketContent = 0;
    int time = 0;
    printf("Time\tPacket Size\tBucket Content\tAction");
    while (time <= totalTime) {
        int currentPacket = rand() % packetSize;
        printf("%d\t%d\t\t%d\t\t", time, currentPacket, bucketContent);
        if (currentPacket <= (bucketSize - bucketContent)) {
            bucketContent += currentPacket;
            printf("Packet added");
        } else {
            printf("Bucket overflow");
        }
```

```
            bucketContent -= inputRate;
            bucketContent = (bucketContent < 0) ? 0 : bucketContent;
            time++;
        }
    }
    int main() {
        int inputRate, bucketSize, packetSize, totalTime;
        printf("Enter the input rate: ");
        scanf("%d", &inputRate);
        printf("Enter the bucket size: ");
        scanf("%d", &bucketSize);
        printf("Enter the packet size: ");
        scanf("%d", &packetSize);
        printf("Enter the total time: ");
        scanf("%d", &totalTime);
        leakyBucket(inputRate, bucketSize, packetSize, totalTime);
        return 0;
    }
    ------------------------
    //Exp 13: Implement data encryption and data decryption.
    #include <stdio.h>
    #include <string.h>
    int main() {
        char text[100];
        int key, i;
        printf("Enter the text: ");
        scanf("%s", text);
        printf("Enter the key: ");
        scanf("%d", &key);
        // Encryption
        for (i = 0; text[i] != '\0'; i++) {
            text[i] = text[i] + key;
        }
        printf("Encrypted text: %s\n", text);
        // Decryption
        for (i = 0; text[i] != '\0'; i++) {
            text[i] = text[i] - key;    }
        printf("Decrypted text: %s\n", text);
        return 0;
    }
    ------------------------------------------------------
    //Exp 14 :: Basic Trouble shooting commands ipconfig,ping ,tracert,nslookup,arp etc.
    ----
    **open command prompt and type the commands.

    1. ipconfig - Displays IP Configuration
    example:
    C:\Users\Pavan>ipconfig
    ------
    2. ping - Checks Network Connectivity
    example:
    C:\Users\Pavan>ping 168.28.1.1 -t (use ur frd ip address)
    ex2 : ping 192.168.1.1 - t
    ---------------
    3. tracert - Traces the Network Route
    example :
    C:\Users\Pavan>tracert mgit.ac.in
    -------
    4. nslookup - Checks DNS Resolution
    example:
    C:\Users\Pavan>nslookup
    Default Server:  UnKnown
    Address:  103.105.177.74

    > mgit.ac.in
    Server:  UnKnown
    Address:  103.105.177.74

    Non-authoritative answer:
```

```
Name:    mgit.ac.in
Address:  65.2.53.139
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5. arp - Displays IP-to-MAC Address Mapping
example:
C:\Users\Pavan>arp -a

Interface: 192.168.10.34 --- 0x10
  Internet Address      Physical Address      Type
  192.168.10.1          b4-f9-49-b9-5e-d6     dynamic
  192.168.10.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
--------------------
```